CAPSTONE PROJECT // 2026-2027

Smart Medicine Box with Robot Assistance

Active scheduling alerts built on an embedded robotic foundation.

Department of Computer Science and Engineering (Data Science) |
St. John College of Engineering & Management

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The Architects Behind the Project



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The Problem: Passive Storage vs. Active Reminders

Ordinary Medicine Boxes.

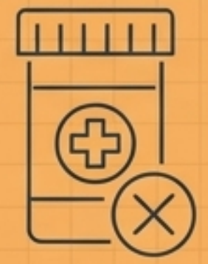
Only store medicines, no active alerts.

Busy Routines.

Distractions cause missed scheduled doses.

Lack of Presence.

The medication is out of sight and out of mind.



Missed Prescriptions.

Taking medicine at the prescribed time is important. We need a system that actively draws attention at the exact scheduled time.

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Literature Review & Technology Gap

Capability	Manual Organisers	Phone Alarms	Electronic Smart Boxes	Proposed Prototype
Physical Separation	✓	✗	✓	✓
Active Audio/Visual Alert	✗	Yes (Depends on user attention)	✓	Yes (Buzzer + LED)
Robotic Foundation	✗	✗	✗	Yes (Integrated hardware chassis)

[!] The practical gap: Integrating scheduled physical reminders with an expandable, low-cost robotic enclosure.

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Objectives & Scope Boundaries

Verified Current Scope (Built Today)

- Assemble a multi-compartment medicine box.
- Trigger an active buzzer & LED for Compartment 1.
- Build a wired robot chassis using an ESP32 and motor hardware.
- Document hardware functions and safe integration paths.

Outside Verified Scope (Built Tomorrow)

- Autonomous, schedule-triggered movement.
- Verification of medicine consumption.
- Mobile notifications & wireless navigation.

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System Architecture

The Reminder Engine (Active)

Medicine Schedule / Controller Logic

Buzzer (existing) | LED Indicator (existing)

User receives audible/visual medicine-time alert

The Robotic Platform (Wired)

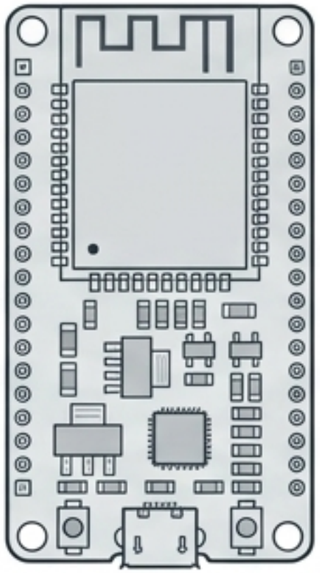
Robot Chassis (Acrylic/3D Printed) + ESP32 + DC Motors + L298N

Wired/Manual Prototype Control

Future Wireless Link + Auto Navigation

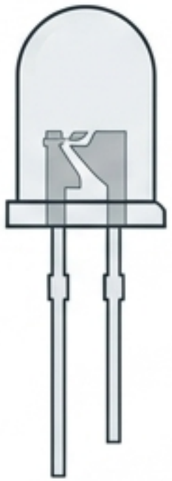
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Hardware Teardown I: The Reminder Engine



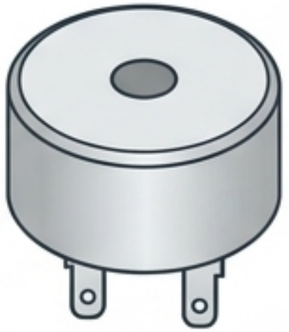
ESP32 Controller

Role: The embedded logic center. Triggers the outputs based on programmed scheduling.



LED Indicator

Role: The visual alert. Blinks when Compartment 1 reaches its configured time.



Piezo Buzzer

Role: The audio alert. Sounds simultaneously with the LED to ensure the reminder is noticed.

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Hardware Teardown II: The Robotic Platform

L298N Motor Driver

Interfaces controller signals with motor current.

12V 300 RPM DC Motors (x2)

Drives the 3D-printed wheels.

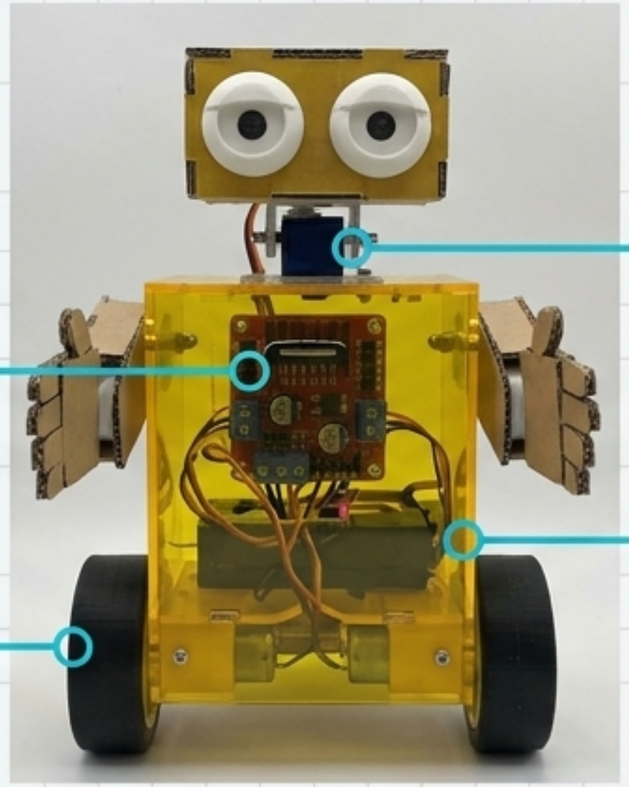


Fig A.2: Front view of the robot enclosure

Mini Servo

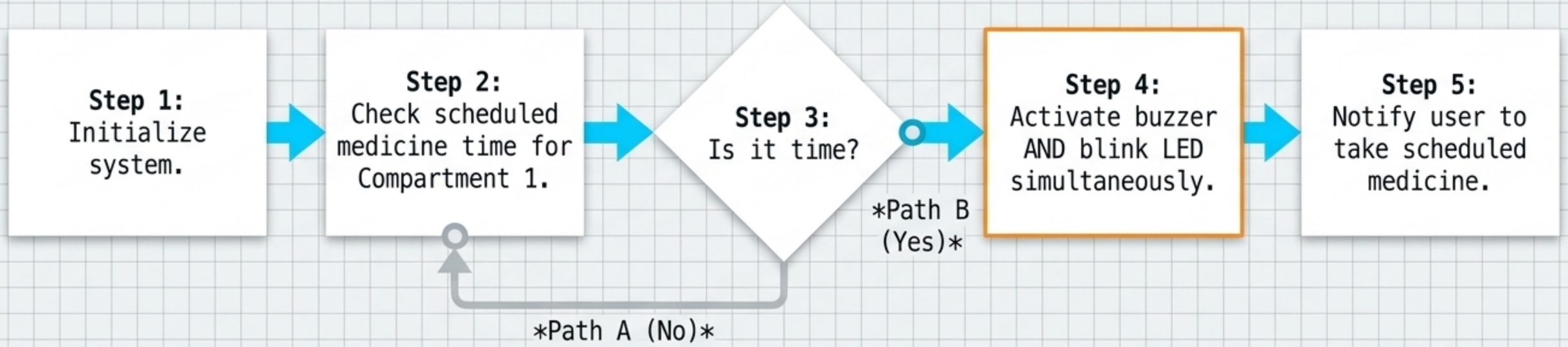
Enables independent neck/head movement.

Chassis Materials

Acrylic body enclosure with yellow masking, combined with 3D-printed custom wheels and face components.

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Current Medicine Reminder Logic



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Project Planning & Execution

Work Breakdown Structure

1. Requirements
Define concept

2. Mechanical
Enclosure, 3D printing

3. Electronics
Controller, motors, indicators

4. Firmware/Testing
Configure logic, test alerts

Team Responsibility Matrix

Om Kadam:	Coordination & Integration
Rishabh:	Assembly & Interface Docs
Pravin:	Controller & Firmware
Om Tripathi:	Testing logs & Diagrams

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Feasibility & Sustainable Impact



SDG 3: Good Health

Designed to support assistive technology demonstrations, making medicine times highly noticeable for users.



SDG 9: Innovation

Built on an affordable, low-cost embedded system (ESP32) that provides a modular foundation for future robotic features.



SDG 12: Responsible Production

Constructed using reusable modules, repairable acrylic/printed parts, and standard components to reduce e-waste.

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Conclusion & Future Scope Roadmap

Phase 1: Current State

Wired Verification. Circuit verification, safe control testing, and active Compartment 1 alerts.

Phase 2: Next Step

Wireless Integration. Implement remote wireless control for the ESP32 and mobile notification integration.

Phase 3: Ultimate Vision

Autonomous Assistance. Schedule-triggered movement, obstacle-aware navigation, and multi-compartment scheduling.

We have established a functional alert system and physical chassis. The foundation is set for full automation.

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